ME240 Spring 2025 Final Report

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Introduction

This report focuses on the ideation, design, testing, and manufacturing processes of creating a pair of bike brake calipers with the engineering goal of minimizing mass and minimizing braking distance while ensuring a smooth stop.

Many steps went into the caliper analysis and design before testing; benchmarking and ISO standards were analyzed in combination with our own engineering analysis to produce a set of needs and metrics to measure the success of the design. Subsequently, the design space was determined, and load magnitudes and application points were specified in conjunction with the acceptable amounts of stress and deflection to better scope the integration of the calipers with the bike. Then, design changes were iteratively made using Finite Element Analysis (FEA) and Topology Optimization (TO) to improve upon our engineering goals.

The calipers were then printed and tested in use on an actual bike during our initial testing session. We then further improved upon the design and printed a final set of calipers for use in our final testing. This report will follow our design process sequentially, noting down all data collected and important observations and discussions of the calculated values.

1. Team Roles

Table 1 lists the team role assignments given to each member of the team at our first meeting. While each person oversaw their respective areas of responsibility, all team members did work in practically every area to ensure smooth workflow and an even distribution of labor.

Table 1: Team role assignments for caliper design project and corresponding responsibilities.

Role	Responsibilities	Member
Scheduling Lead	Scheduling team meetings, ensuring deadlines are met, and overseeing the submission of all assignments	Naomi Cohn
FEA and Topology Lead	Ensures computational simulations are logical and that all team members are capable using these techniques	Sophia Wang
CAD Lead	Ensures that the completed part can be manufactured effectively and engineering drawings and models are accurate	Olivia Petrillo
Manufacturing Lead		Melanie Kaplan
Documentation Lead	Ensures design choices taken by the group are well documented and clearly explained at each step in the design process	Crystal Cooper

2. Needs and Metrics Tables

Needs and metrics were determined before modeling to clearly outline our design and testing objectives. The team held multiple brainstorming sessions to come up with an extensive list of design needs, which is shown in Table 2. As part of this process, we consulted relevant engineering standards such as ISO 4210 and general safety standards and precautions. We also considered user experience, manufacturability, and the given design constraints. We wanted the calipers to not only be functional, but also practical and a commercially viable market competitor.

Each need was assigned a value between 1-5 that established its importance in our final design; the scale of importance was determined based on user experience and safety, along with project requirements, and the scale is defined in Table 3. These rankings document the priority of the needs, which helped to facilitate design choices when multiple needs conflicted and to decide which aspects were most important to focus on improving.

Table 2. Design Needs for Brake Calipers and Ranking of their Importance in Caliper Design.

#	Need		Importance
1	The caliper	slows the rotation of the bike wheel	5
2	The caliper	integrates easily with the provided mounting hardware	5
3	The caliper	is aesthetically pleasing	1
4	The caliper	can be maintained with readily available tools	2
5	The caliper	lasts a long time (test cycles)	3
6	The caliper	minimizes weight	5
7	The caliper	is affordable to produce	3
8	The caliper	is safe in a crash	4
9	The caliper	decelerates the bike in dry and wet conditions (acceptable ratio)	4
10	The caliper	decelerates the bike quickly (acceptable distance, range)	5
11	The caliper	decelerates the bike safely (range, works with rear and both, a “margin of error so to speak”)	4
12	The caliper	is sized to not impede use (pedaling or sitting)	3
13	The caliper	is compliant with ISO standards 4210-2 and 4210-4	5
14	The caliper	is made of non-toxic materials	4
15	The caliper	maintains position during use (uses locking devices for all screws (lock washers, lock-nuts, threads, stiff nuts) Cable ends are protected to prevent unraveling (cap or treatment)	4

16	The caliper	decelerates the bike with smooth safe-stop characteristics: no excessive juddering, front wheel locking, bicycle overturning, rider's loss of control, or excessive side skid causing the rider to put their foot to the ground to retain control	4
17	The caliper	is manufactured using Nylon-12 on a Formlabs Fuse 1+ Printer	5
18	The caliper	is comfortable to use (force required) stops bike with reasonable force	3
19	The caliper	is durable and resistant to deformation	3

Table 3: Design Needs Importance Ranking Scale and Justification.

Ranking Scale	Description
1	Not very important, little impact on performance or user experience
2	Somewhat important, some impact on performance or user experience
3	Moderately important, impacts performance or user experience
4	Very important, impacts the safety of use
5	Explicitly asked for in project requirements, or core functionality of the device

The team then determined metrics, which are outlined in Table 4, to measure the success of the caliper design in meeting the needs listed above. Methods for measuring these metrics included physical measurements and Likert scales based on opinion. All needs are described by at least one metric, with some covering multiple needs at once. We also had to ensure that we could easily perform the measurement of these metrics once we reached the testing stage. This meant that no extensive fatigue testing could be performed, despite its relevance to the long-term and consistent use case of typical brake calipers. Ultimately, ease and speed of measurement took precedence to streamline our design process and be able to rapidly improve and test our designs.

Table 4. Design metrics and testing for brake calipers, including which needs they address and what method and units were used to measure them.

#	Needs Addressed	Metric	Units	Method
1	1, 13	Change in Angular Velocity of the wheel before and after braking for 3 seconds.	rotations per second, or Hz	User testing

2	10, 11, 13	Braking Distance (Dry, initial velocity is 25 km/hr, rear brake only)	Meters	User-testing
3	10, 11, 13	Braking Distance (Wet, initial velocity is 16 km/hr, rear brake only)	Meters	User-testing
4	9, 13	Ratio between wet and dry braking distances taking into account the different test velocities.	Ratio	User testing/ standards
5	6	Brake Caliper Mass	grams	Benchmarking
6	18, 13	Input operating force	Newtons	User testing
7	15	Sustainable impact force	Newtons	Engineering Analysis
8	15	% change in brake pad-tire gap (before and after all user-testing)	Percentage	User-testing
9	5, 13, 19	Security Test (ISO 4210-4: 2023 4.3)	Breaking repetitions without failure	User-testing
10	14, 13	Solubility testing of ASTM F963 elements within materials and coatings	ppm	Regulatory Standards
11	17	Compliance verification with material constraint of Nylon-12 on a Formlabs Fuse 1+ Printer	Binary Metric	User testing
12	7	Cost	Dollars	Benchmarking
13	12, 3	Width of brake caliper	Millimeters	Benchmarking

14	15, 8, 13	Attachment of brake assembly and cable requirements (cable ends covered)	Binary Metric	User testing
15	4	Likert scale: Have users assemble and disassemble using basic tools that would be in a typical toolbox set and rank ease	Scale from 1-5	User testing/ Benchmarking
16	8	Likert scale: Ask users who felt and operated the brake calipers how safe they feel and appear	Scale from 1-5	User testing
17	2	Likert scale: Asks assemblers how easily the brake caliper integrates with mounting hardware such as brake pads and cables.	Scale from 1-5	User testing/ Benchmarking
18	3	Likert scale: Ask users on a scale of 1-5 how aesthetically pleasing they find the brake caliper	Scale from 1-5	User testing/ Benchmarking
19	16, 13	Likert scale: On a scale from 1-5 how smooth was the deceleration using the brakes	Scale from 1-5	User testing

For each metric, we also determined an ideal value and a marginal value, which are outlined in Table 5. The ideal and marginal values act as a range of acceptable values, where the marginal is the lowest acceptable value to maintain the integrity of the calipers, and the ideal is the reach value. We strived to hit the ideal values when developing our design, and prioritized optimizing metrics for the needs of greater importance ranking. All marginal and ideal values are justified through ISO standards, research, or benchmarking to ensure credibility. Caliper Benchmarking is outlined in Appendix A.

Table 5. Ideal and marginal values for brake caliper design metrics with justification of the chosen values based on ISO Standards, benchmarking, and engineering analysis.

#	Ideal Value	Ideal Value Reasoning/Source	Marginal Value	Marginal Value Reasoning/Source
1	0 rpm after breaking	Ideally, the brake would stop the rotation of the bike completely within 3 seconds for a normal riding speed, which is a value referenced in ISO 4210-4: 2023 4.6.3.5	A final angular velocity 50% smaller than the initial	The brake needs to significantly slow the rotation of the bike wheel over time, and slowing of the bike wheel is essential for further testing of braking distance, braking time, and other tests used in ISO 4210-4.
2	10 m	Ideally would have a braking distance of 2/3 the upper limit to balance quick stopping with the rider's inertia to create a comfortable deceleration	15m	Value from ISO 4210-2:2023 4.6.8.1.2 Table 1, Testing parameters described in ISO 4210-4 2023: 4.6.3
3	6.6 m	2/3 of the upper limit	10m	Value from ISO 4210-2:2023 4.6.8.1.2 Table 1, Testing parameters described in ISO 4210-4 2023: 4.6.3
4	1	Ideally the performance would be unaffected by wet conditions and would perform equally well in both cases. A ratio of 1 means they decelerate at the same rate.	4:10 (wet to dry)	Ratio calculated using formula 2 from ISO 4210-4:2023 item 4.6.3.11 c) and using braking distances from Table 1 in ISO 4210-2:2023
5	100 g	See Appendix A (Caliper Benchmarking)	190 g	See Appendix A (Caliper Benchmarking)
6	90 N	50% of the maximum required force to brake according to ISO standards.	180 N	Maximum force allowed in ISO 4210-4:2023: 4.3 security test to see if brake works
7	900 N	200% of the impact force of a typical bike crash	450 N	Impact force (F= 2mv/t) of average bike crash going 10 mph= 4.47 m/s, with 100 kg total mass and crash time of 2 seconds. Source: Helmets.org

8	0%	Ideally, would not change at all	10%	Must still be a significant distance between the brake pads and the wheel
9	Infinite	Ideally would be able to perform infinite repetitions without failure, both forwards and backwards	5	Required repetitions used in ISO 4210-4: 2023 4.3 Security Test
10	0 ppm	Ideally there would be no presence of any of the ASTM F963 elements within our design materials so there is zero possible exposure	Solubility limits set forth in Table 1 of 16 CFR Part 1253	Solubility limits in Table 1 are governmental standards for all consumer goods used by children Source: Govinfo.gov
11	Yes	Part should ideally be made out of Nylon-12 on a Formlabs Fuse 1+ printer per instructions.	Yes	Part must be made out of Nylon-12 on a Formlabs Fuse 1+ Printer per instructions Source: Lecture 1
12	$17	80% cost cut from average brake caliper price. See Appendix A.	$87	See Appendix A (Caliper Benchmarking)
13	100 mm	Average width of top of bike fork. Ideally for aesthetic and usability reasons, the brake calipers should not protrude past the frame of the bike. Source: Seraphbikes	148 mm	Average Q-factor, or the lateral distance across the two crank arms between their outer edges, for a road bike. Our caliper should not have a width larger than this factor to allow for proper biking technique. Source: ICAN cycling
14	Yes	Must meet requirements outlined in ISO 4210-2 20203: 4.6.3	Yes	Meets requirements outlined in ISO 4210-2 20203: 4.6.3
15	5	Easy to assemble and disassemble, entirely possible with basic tools	3	Somewhat simple to assemble and disassemble, mostly possible with basic tools as benchmarked against other brakes

16	5	No physical or visual safety issues, no exposed edges that come into contact with user during normal handling and maintenance	3	Appears safe enough, with small protruding edges but all rounded, capped or processed.
17	5	Integrates seamlessly with no issues, even enhancing functionality	3	Integrates ok as benchmarked against other brake calipers
18	5	Very aesthetic compared to other commercially available brake calipers	3	Average aesthetic benchmarked against other brake calipers
19	5	Use of brake exhibited all smooth safe-stop characteristics outlined in ISO 4210-2 2023: 4.6.8.2	3	Use of brake exhibited most smooth safe-stop characteristics outlined in ISO 4210-2 2023: 4.6.8.2

3. Design Concept Development

1. Estimation of the Design Space

To develop the caliper design, we first needed to understand our design space, or the area in the assembly that we own and could add material to. To estimate the design space, we started by looking at the bicycle assembly we would model our calipers around, as shown in Figure 1. From this, we determined the basic shape of the space available and identified connection points.

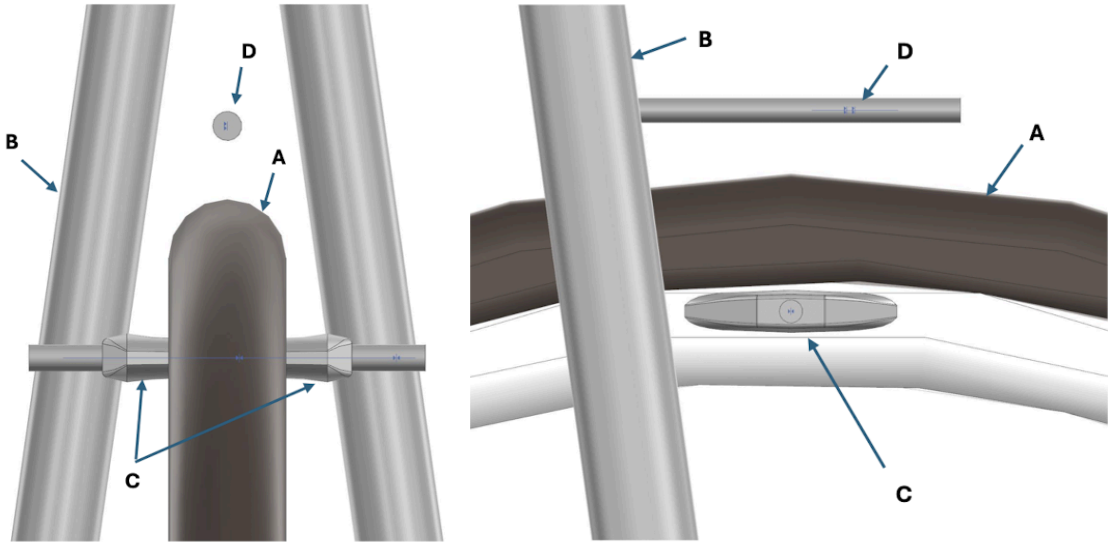


Figure 1: Front and right views of the bicycle assembly used to determine design space with the (A) wheel, (B) bike frame, (C) brake pads, and (D) pivot point.

We knew each caliper needed to be secured onto the pivot point and have a brake pad attached to it. The parts could not touch the wheel or frame either, and we needed to account for a brake cable connection connecting both calipers. Knowing these parameters, we generated three orthographic sketches that captured everything we had gathered, as shown in Figures 2, 3, and 4.

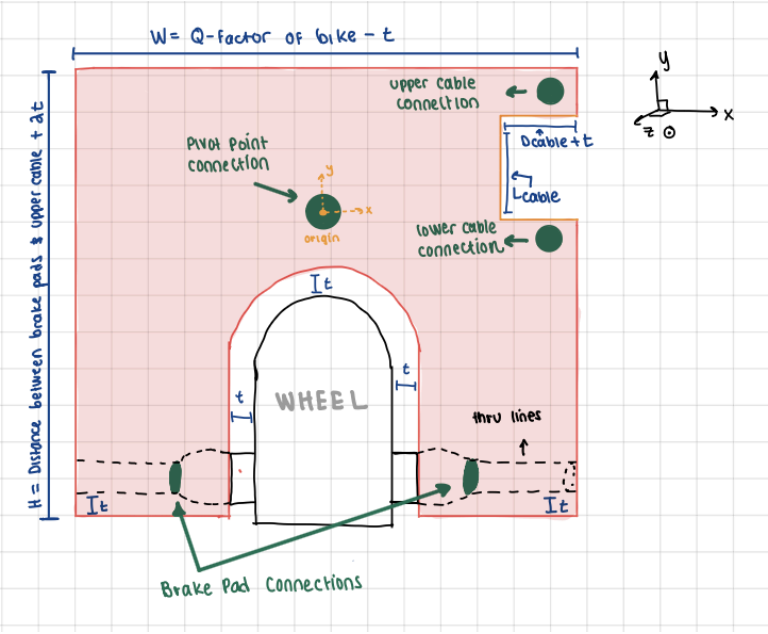


Figure 2: Front view of the 2D design space with dimensional labels.

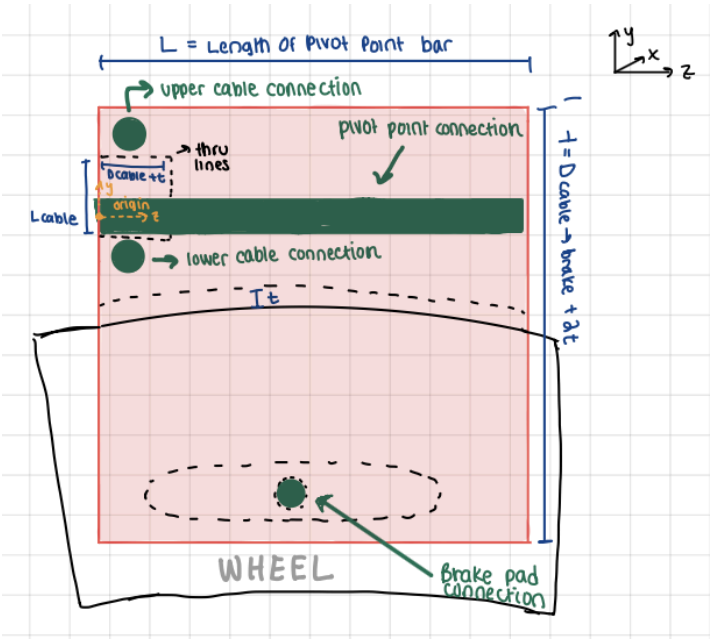


Figure 3: Right view of the 2D design space with dimensional labels.

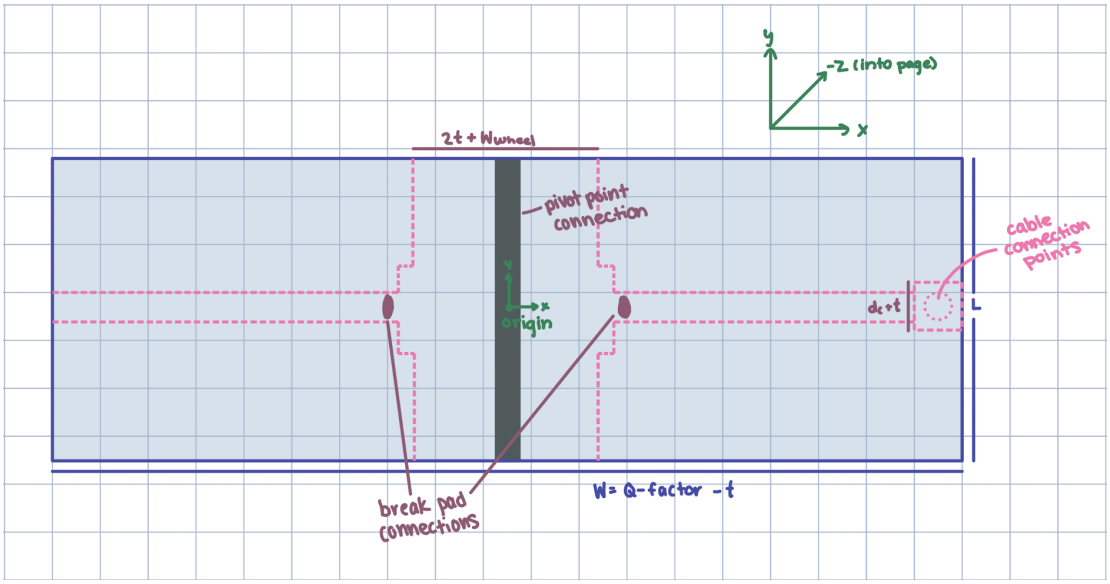


Figure 4: Top view of the 2D design space with dimensional labels.

Each of the specific dimensions we chose for the design space are described in Table 6 below. These parameters were chosen based upon benchmarking and research on brake sizes in industry.

Table 6: Legend for design space sketch parameters and a description of how they were determined.

Variable Name	Meaning
t	A small tolerance to ensure design space does not intrude on free movement or the integrity of other components
H	Height of the design space that includes all necessary design components, the largest vertical distance between the connection points
W	Width of the design space, chosen as the Q-factor of the bike, which is the horizontal distance between the outer faces of the crank arms
D_{cable}	Diameter of cable with tolerance
L_{cable}	Length of cable between two connection points
L	Length of pivot point bar
W_{wheel}	Width of the wheel
Dashed lines	Indicate design space features below the surface level.

After creating these preliminary 2D sketches, we moved on to a 3D model of the design space, which is illustrated in Figure 5 below. The model captured each of the dimensions assigned to

the orthographic sketches. When making said 3D model, we also split the design space into two pieces for the left and right calipers. The location of this split was also based on benchmarking and research. We made sure to leave room for every external component that would need to be connected to the calipers.

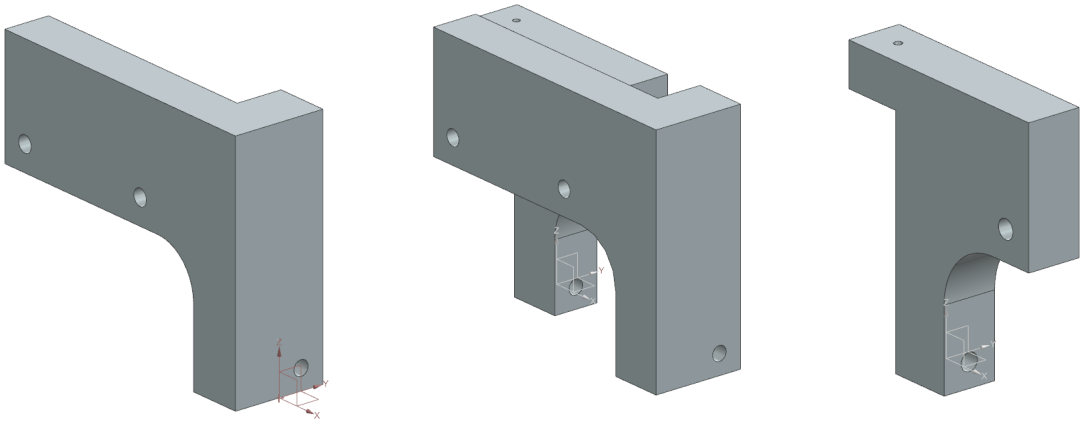


Figure 5: Trimetric views of the 3D modeled design space with the space split into two separate design spaces, one for each caliper.

2. Concept Sketches

With our design space fully defined, we then moved on to creating initial designs. For our original concept sketches, we decided to utilize benchmarking and base them on bicycles around campus. We identified 4 different bikes with rim brakes and took some photos, which are shown in Figure 6. These designs were all very thin and rounded and made out of metal. We noted that our design would have to be slightly different, as Nylon-12 is a less strong material.



Figure 6. Brake calipers from bikes around campus we used for initial design concepts as part of our benchmarking analysis.

Taking into account the general shape of each of these calipers, we then made a few sketches with varying amounts of roundness, which are shown in Figure 7. The first sketch was most similar to the benchmarked brakes, but the thin structure did not seem like it would be stiff enough if made from Nylon-12. We then tried adding thickness to our designs while still using the same general shape, and then finally creating a simplified version with sharp defined edges.

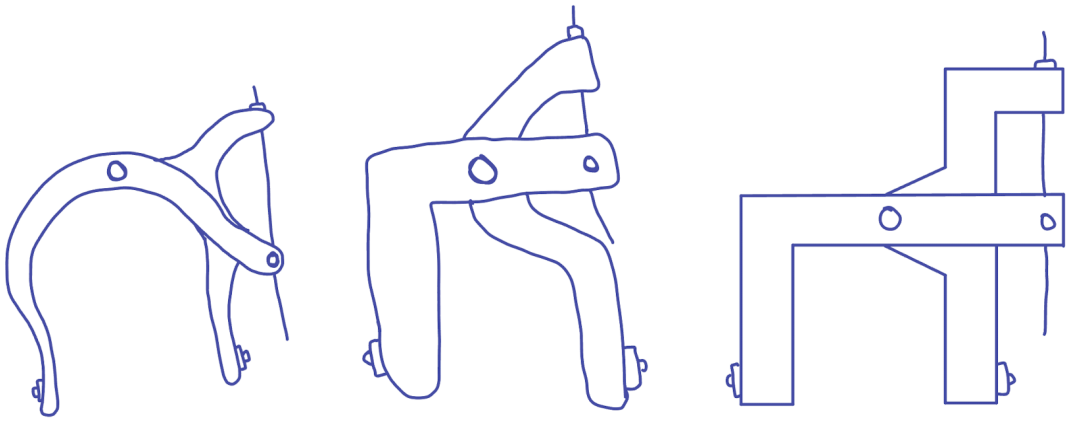


Figure 7. Three different concept sketches created based on our benchmarking of campus bikes. They vary from left to right with thickness and extent of roundness.

These first two sketches were slightly trivial. We decided to work off of the sketch on the right in Figure 7 to create our solid model, as it has a similar geometry and the dimensions could be much more easily changed. Starting with simpler and more robust geometry streamlined the process of creating iterative design changes. Our initial models are shown below in Figure 8.

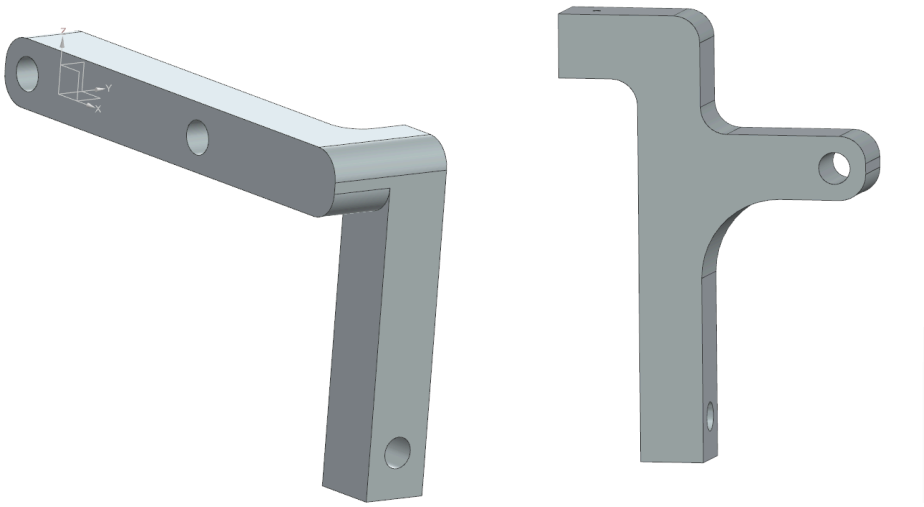


Figure 8: Initial right and left caliper solid models, created from initial design sketches.

3. Optimization of Load Paths

Before doing topology optimization, we used engineering analysis and knowledge of load path behavior to predict load path locations and optimize our design to follow them. Load paths in our calipers describe how loads will travel through our calipers to meet other forces. Loads tend to travel in straight lines, or mild arcs, over the shortest available distance.

There are three types of forces acting on each caliper that will create load paths: the tension force from the brake cable acting at the brake cable mount hole, the normal force from contact with the tire acting at the brake pad mount hole, and the x and y reaction forces acting at the pivot hole, which acts as a pinned constraint on the system. These forces were mapped onto our design space, acting on the connection points. Then, load paths were drawn according to the following rules, as shown below in Figures 9 and 10:

1. The load paths must remain in the design space, because that is the only possible location where material can be added.
2. The load path acts in the straightest possible line between the locations of the applied forces.

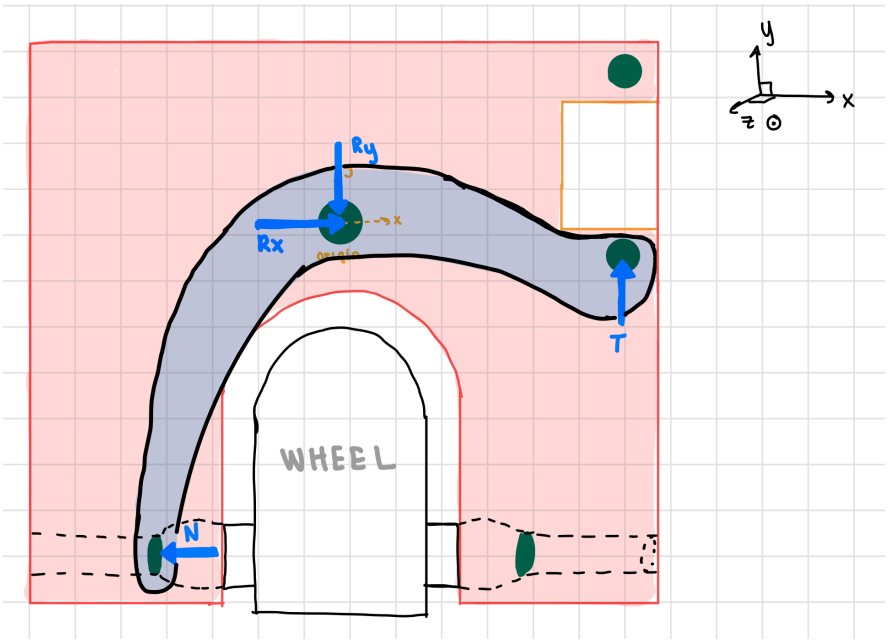


Figure 9: Right caliper free body diagram and load path optimization sketch.

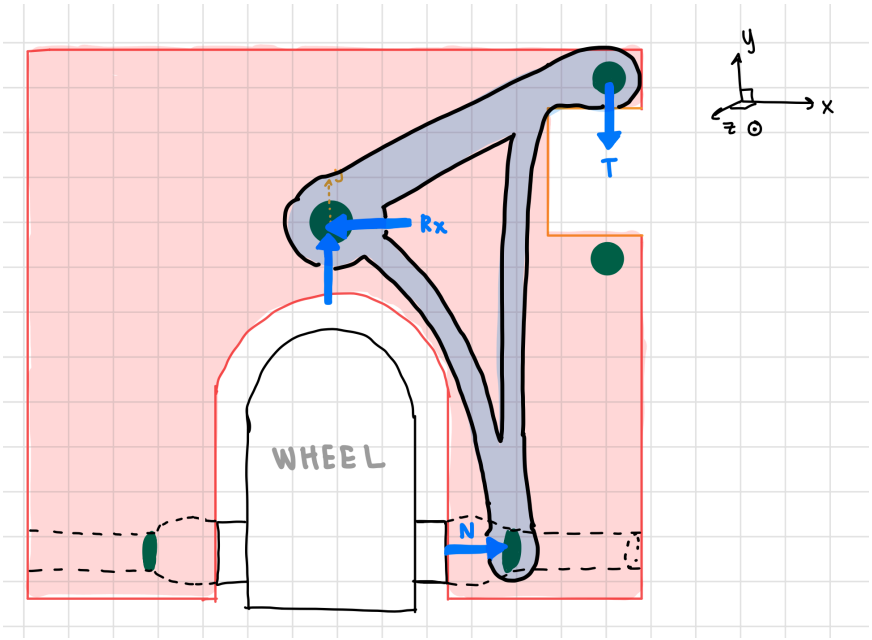


Figure 10: Left caliper free body diagram and load path optimization sketch.

We further optimized our load type sketches based on the types of loads and the stress response to them. Both calipers are subjected to combined loads, and we know that stresses resulting from bending loads tend to be much higher than stresses from other types of applied loads. Therefore, we predicted that more material would be needed when load points are located further apart, specifically in sections of the caliper between the pivot point and the brake cable mount points.

Finally, we acknowledge that our design needed to mitigate stress concentrations, which occur when load paths concentrate within a part due to sudden changes in the cross-sectional area. Thus we planned to optimize our design for load paths by avoiding sharp corners and creating gradual changes in cross-sections where applicable. This idea is especially important to the design of our calipers, since they are made from the relatively brittle material Nylon-12. Load paths are further analyzed through topology optimization later in the design process and are discussed in Section 3.5.

4. Hand calculations

Engineering analysis was used to calculate the values of the applied loads, as well as the maximum allowable values for deflection and stress on the calipers. These values were used when constructing and interpreting FEA and TO results.

Applied Loads:

As stated in Section 3.3, three forces were identified to be acting on each brake caliper: a tension force from the brake cable, a normal force from contact with the tire at the brake pad mount, and the reaction force within the pivot point, which acts as a pinned support. The magnitudes of these forces were found through hand calculations.

Initially, we incorrectly calculated these forces by starting with a hand brake input value of 180 N, which we then converted into a tension cable force through moment equations at the handle. This resulted in a tension force of 720 N, which we identified as alarmingly large. We realized our approach was wrong, since the force to "bottom out" our brakes would vary with the stiffness of our calipers, and would not necessarily ever reach 180N.

We then re-worked our approach and started with concretely known values from the ISO standards brake test, namely the mass of the rider and bike and the velocity of the bike before braking, and the maximum acceptable braking distance for that speed. We then used energy conservation to arrive at the equation

$$\frac{1}{2}mv^2 = F_{fr}d_{br},$$

This question says that all of the kinetic energy of the bike before braking must be equal to the work done by the brake to bring the bike to a complete stop. The work is the product of the frictional force of the tire with the ground and the distance it takes to brake. Substituting in our values from the ISO standards, we were able to calculate the force of friction.

We then measured the distance from the wheel axis to both the brake pad and the outer edge of the tire, and used these values in a moment balance equation to convert the frictional force of the tire against the ground to the frictional force of the brake pad against the tire.

$$F_{fr}r_{tire} = F_{brf}r_{br}$$

The provided frictional coefficient of 0.9 of the brake pad was used to convert this frictional force to the normal force between the brake pad and the tire using a known physical relationship.

$$F_{brf} = \mu N = 0.9N$$

Finally, the vertical distance between the brake mount point and the pivot point was measured and labeled as r_{pad}, and the horizontal distance between the cable mount point was measured and labeled as r_{cable}. These two distances were used to create a moment balance equation around the pivot point, which was used to calculate the tension force in the brake cable.

$$Nr_{pad} = Tr_{cable}$$

As our design changed, we altered the values for r_{pad} and r_{cable} and then adjusted the applied forces accordingly. All of these calculations were done in a google sheet, which is condensed and shown in Appendix B.

Maximum Allowable Stress:

The following values were calculated while analyzing stress. Since these calipers will be loaded cyclically, we could not design the caliper with just the ultimate tensile strength of the material.

1. Ultimate Tensile Strength of SLS-Nylon-12: 50 MPa
2. The endurance strength for 10,950 cycles to failure: 36.5 MPa
3. Safety Factor: 1.3
4. Maximum allowable stress: 28.07 MPa

In order to ensure our factor of safety and maximum stresses were acceptable, we calculated the endurance strength of Nylon 12 over 10,950 cycles. 10,950 was chosen as the number of cycles based on estimations about biking conditions and material lifetime for everyday use for 1 year, and resulted in an endurance strength of 36.5 MPa, as seen in Appendix C. Applying our desired safety factor of 1.3, the maximum allowable stress of our part is 28.07 MPa.

Maximum Allowable Deflection:
The following values were used while analyzing the deflection values of our brake calipers:

1. Allowable deflection per design: 11 mm
2. Safety Factor: 4
3. Maximum total allowable deflection: 2.75 mm

The brake calipers were designed with an 11 mm distance between the right and left connections to the brake cable, which means that 11 mm is the maximum total deflection that the calipers can incur before failure occurs. We chose a factor of safety of 4 to ensure that this critical requirement was met and to ensure safe use. FEA often underestimates deflection, so we have increased uncertainty from our stress calculations, which is why this factor of safety is larger than it was for stress. Applying our desired safety factor of 4, the maximum total allowable deflection summed between both calipers is 2.75 mm.

5. Topology Optimization

We used topology optimization to optimize our calipers and determine what locations needed more material than others. We ran two topology iterations on both the right and left design spaces using different objectives. For our first run, we ran TO according to the given lab instructions, which were to reduce mass to 25% of the original design space and to maximize stiffness. We used the input loads calculated in 3.3 above. The TO results are shown in Figures 11 and 12.

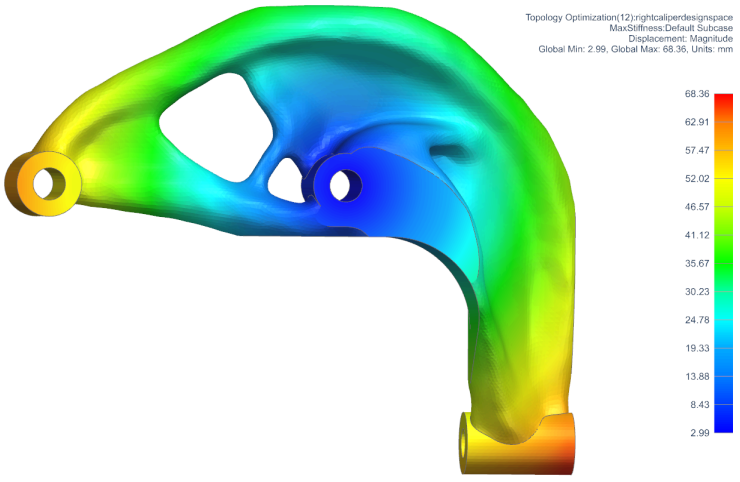


Figure 11: Right brake caliper topology optimization with "Maximize Stiffness" design objective and optimization constraint of 0.25x original mass of design space.

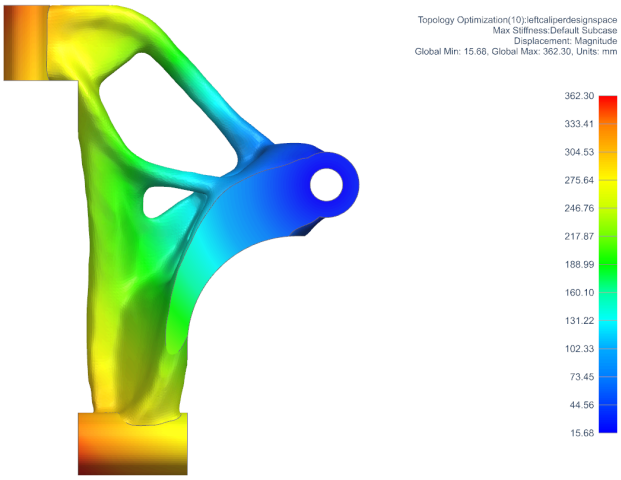


Figure 12: Left brake caliper topology optimization with "Maximize Stiffness" design objective and optimization constraint of 0.25x original mass of design space.

However, we realized that the lab instruction setup did not necessarily align with our desired metrics, so we ran a second round of TO with different settings. This time we changed the initial optimization objective to "Minimize Mass" and set our optimization constraint to have a maximum deflection of 2.75 mm, which represents the maximum deflection multiplied by a factor of safety of 2. We also then added both a pinned analysis constraint at the pivot hole and a planar slider analysis constraint at a small cylindrical extrusion representing the brake pad to fully constrain our model and get rid of the error message seen in the first run. The results of the second round of TO are shown in Figures 13 and 14 below.

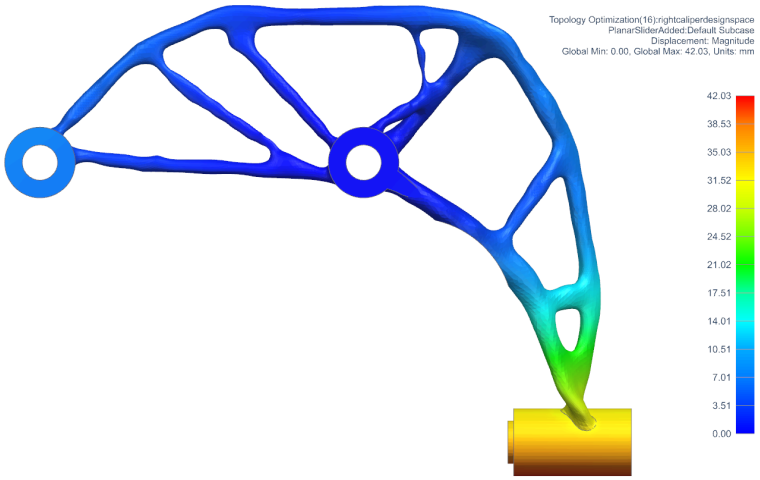


Figure 13: Right brake caliper topology optimization with "Minimize Mass" design objective and optimization constraint of a deflection of 2.75 mm (Factor of Safety of 2.0 for 5.5 mm allowable deflection), with planar slider analysis constraint added.

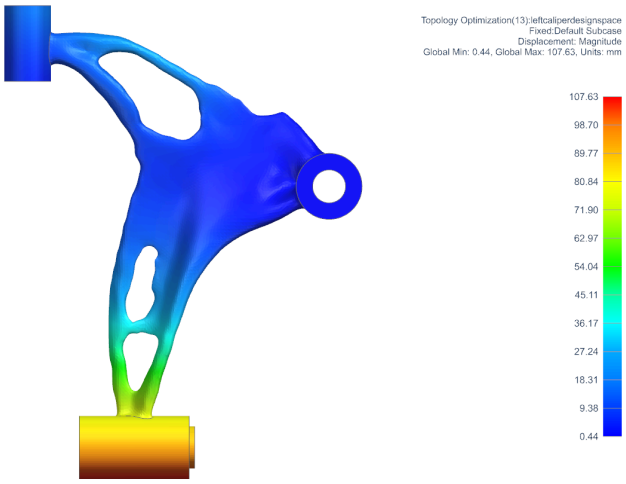


Figure 14: Left brake caliper topology optimization with "Minimize Mass" design objective and optimization constraint of a deflection of 2.75 mm (Factor of Safety of 2.0 for 5.5 mm allowable deflection), with planar slider analysis constraint added.

Both rounds of TO resulted in common features, specifically the reduction of material within the center of the bodies. We also realized that we could remove material on the corners of the calipers and change the shape of our calipers to be more curved. Our initial caliper models were altered to reflect these observations. Figures 15 and 16 below show the changes made to the right and left caliper designs as a result of observations from running TO. Further views of the updated calipers are shown in Figures 17 and 18.

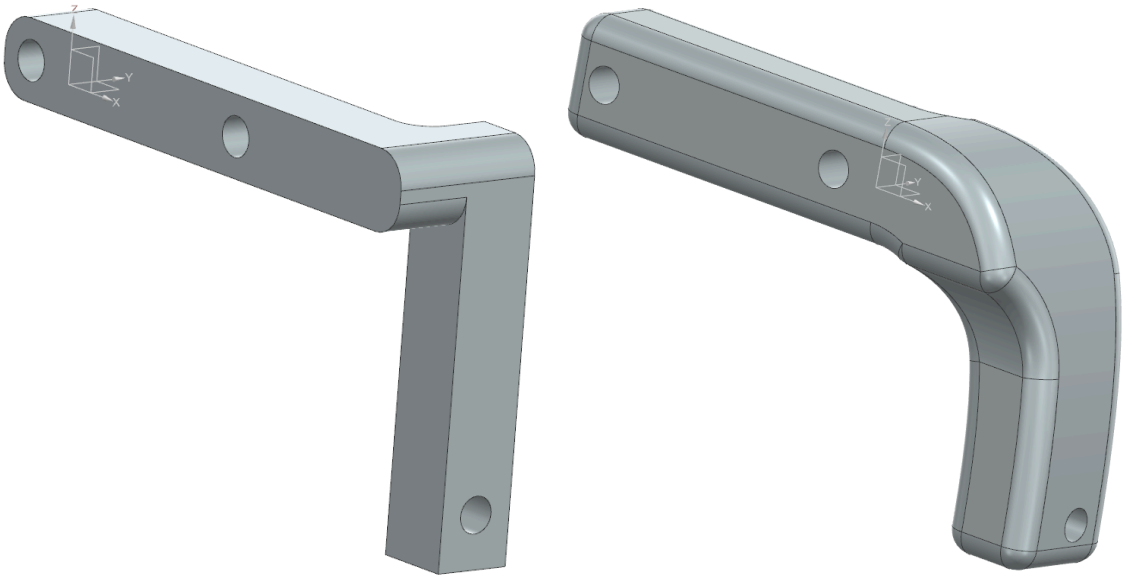


Figure 15: Design changes made to the right caliper as a result of Topology Optimization. TO suggested a rounder design with more material vertically, as well as more rounded edges.

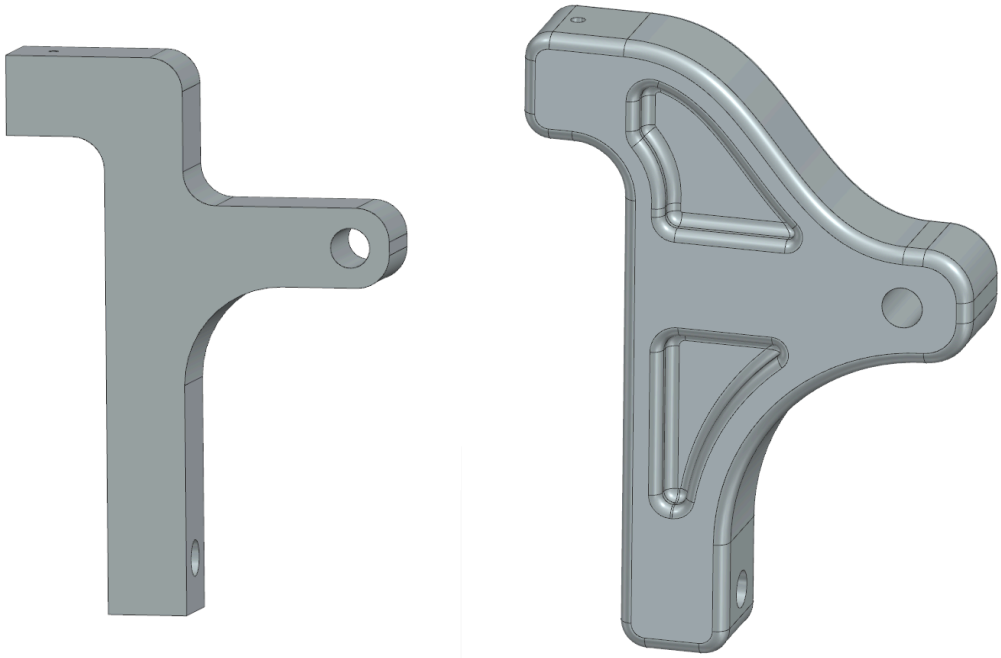


Figure 16: Design changes made to the left caliper as a result of Topology Optimization. TO suggested a rounder design with more direct load paths between the pivot, mounting, and brake holes, as well as the ability to reduce weight by subtracting some material from the inside.

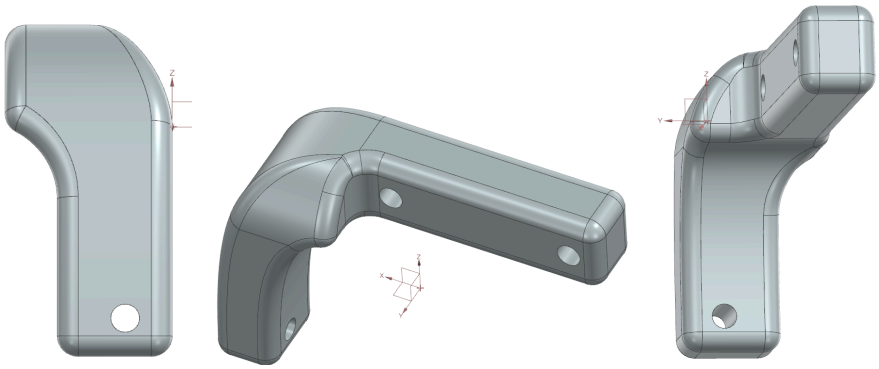


Figure 17: Various views of the right caliper for the TO updated design.

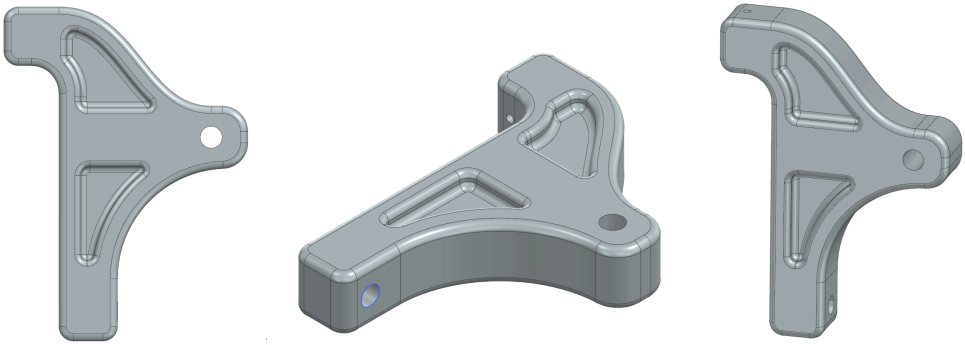


Figure 18: Various views of the left caliper for the TO updated design.

6. Iterative Usage of Finite Element Analysis (FEA)

A large portion of the design process was spent performing iterative FEA on each caliper to determine the effects of design changes. We first performed FEA on our initial caliper designs as a baseline and then compared this to the updated caliper designs to document changes in stress values.

The Von-Mises stress of the initial and updated right caliper designs are seen in Figures 19 and 20 respectively. The maximum stress on the original right caliper is located at the inner corner of the connection from the vertical and horizontal components and is about 150MPa. The maximum stress on the improved right caliper is 21 MPa in about the same location. Per the directions, we are ignoring the high stresses around the brake pad connection point and pivot point, which come from the way we set up the FEA. With a maximum allowable stress of 22.8 MPa using our factor of safety of 1.6, we see that the updated right caliper's maximum stress is below this value and meets our design needs.

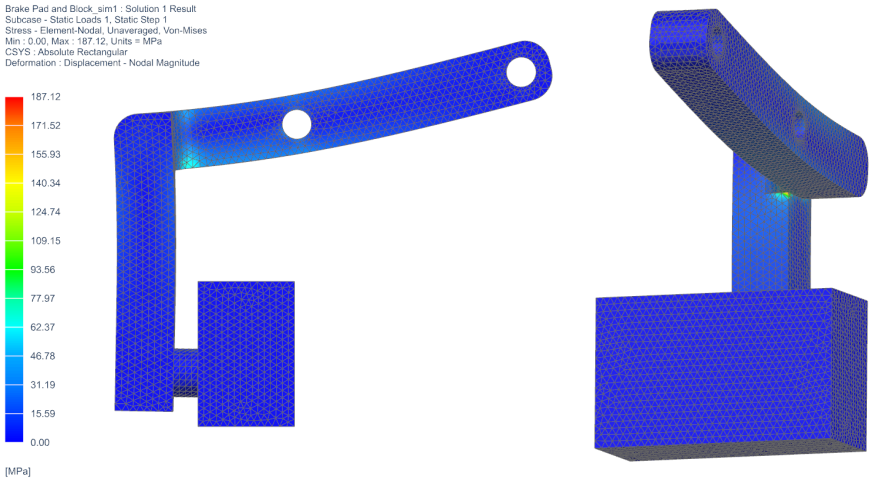


Figure 19: The calculated Von-Mises stress on the initial right caliper model from FEA.

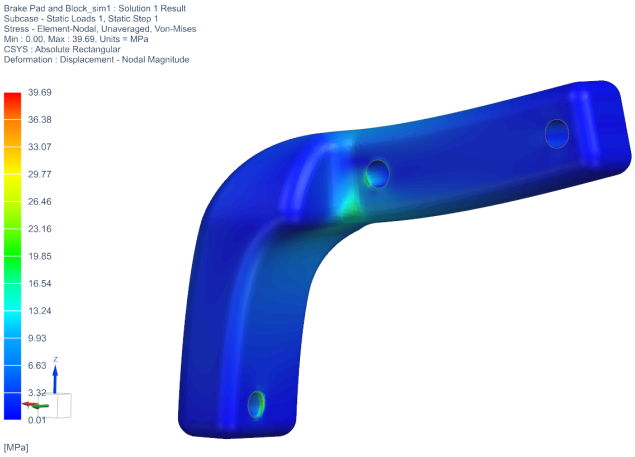


Figure 20: The calculated Von Mises stress on the updated right caliper model from FEA.

The Von-Mises stress of the initial and updated left caliper designs are seen in Figures 21 and 22 respectively. The maximum Von Mises stress on our original caliper was about 85MPa, as shown in the orange-yellow corner of Figure 21. After we improved our calipers using feedback from the topology optimization, the new maximum stress value is 21.63 MPa and is located in the corner of the bottom cut-out. We are able to ignore the maximum recorded stresses at the pivot hole, as this is an artifact of how we constrained this hole as a pinned support. With a maximum allowable stress of 22.8 MPa using our factor of safety of 1.6, we see that the left caliper's maximum stress is also below this value and meets our design needs.

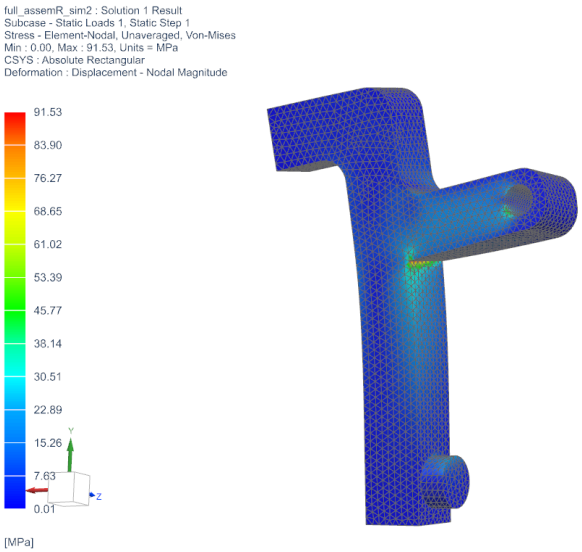


Figure 21: The calculated Von-Mises stress on the initial left caliper model from FEA.

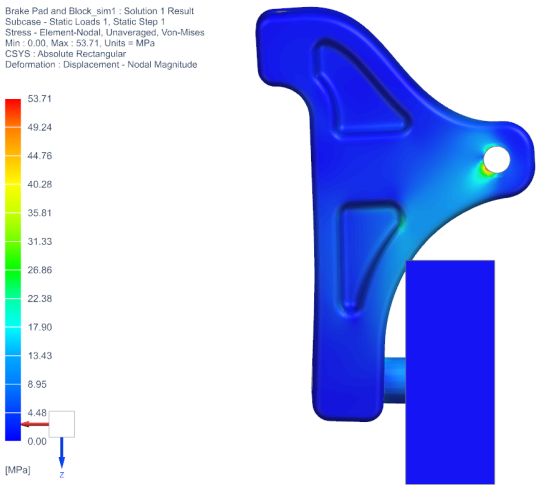


Figure 22: The calculated Von Mises stress on the updated left caliper model from FEA.

The deflection of the initial and updated right caliper designs are shown in Figures 23 and 24 below. The maximum deflection of our original right caliper was 9.35 mm; however, upon

improving our caliper designs using insights from the TO, we were able to decrease the deflection to a maximum value of 1.474 mm in the vertical direction.

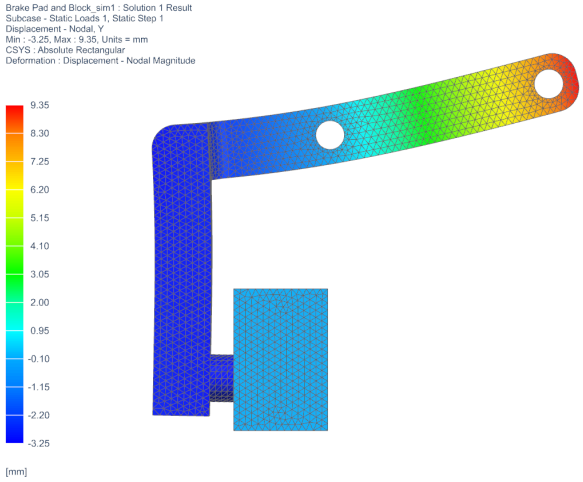


Fig. 23: FEA output for Y nodal (vertical) deflection of the initial right caliper design.

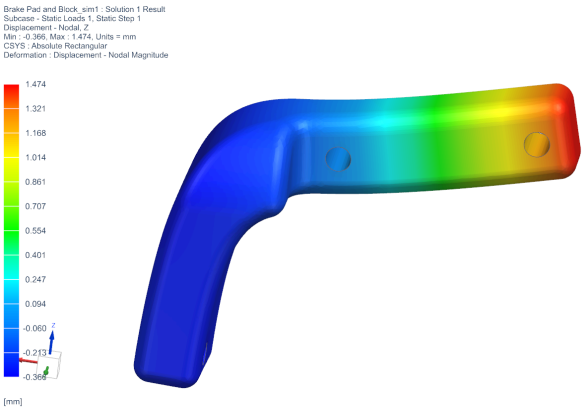


Fig. 24: FEA output for Z nodal (vertical) deflection of the updated right caliper design.

The deflection of the initial and updated left caliper designs are shown in Figures 25 and 26 below. The maximum deflection of our original left caliper was 9.35 mm; however, upon improving our calipers with the TO, we were able to decrease it to a maximum of 1.474 mm in the vertical direction.

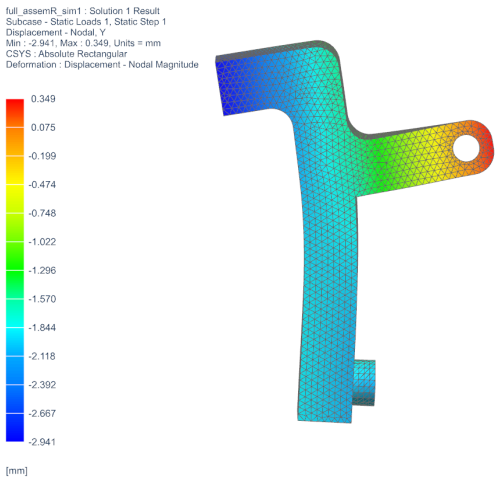


Fig. 25: FEA output for Y nodal (vertical) deflection of the initial left caliper design.

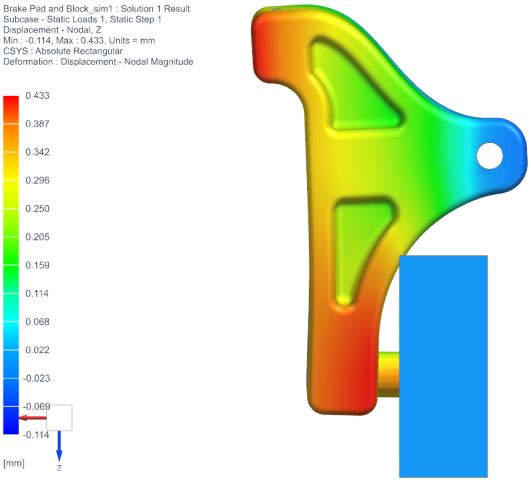


Fig. 26: FEA output for Y nodal (vertical) deflection of the updated left caliper design.

The total deflection of the initial caliper design exceeded the total allowable deflection. However, the total deflection of the updated caliper design is 1.904 mm. Therefore, we can conclude that our updated caliper design meets our maximum allowable deflection requirement with a factor of safety of 4.

4. Initial Testing Results

The updated caliper designs analyzed in Section 3.6 were printed and tested for initial caliper testing. This design successfully passed all of the design metrics. The results of the initial caliper testing are shown in Table 7 below.

Color Key:

Did not meet metric	Met marginal value	Met ideal value

Table 7: Design metrics for brake caliper design specified with units and how it was measured. Also included are the ideal and marginal value, and the outcomes from initial testing.

Metric	Marginal	Ideal	Measured	Units	Measurement Tool
Change in angular velocity before and after braking for 3 seconds	50% of initial angular velocity	0	0- braked for exactly 2.9 seconds	rpm	Video Freeze Frame
Braking distance (dry, initial velocity 25 km/hr)	15	10	15.0 m	meters	Measuring tape, Velocity meter, Equation 4 Lab 8 pdf
Mass	190	100	87.5 ± .5 g (50.5 g right, 37 g left)	grams	Scale
Input operating force	180	90	120 N (maximum applied force, bottomed out)	N	Force Gauge
Caliper Width	148	100	90.91 mm	mm	Caliper
% change in brake-tire gap	10	0	0	%	User testing, by inspection
Cable Assembly (cable ends covered)	yes	yes	yes	Binary metric	User testing, by inspection
Ease of Assembly and disassembly using basic tools	3	4.5	3 (Keys)	Scale 1-5	User testing (Likert scale)
Caliper's safety based on feel & appearance	3	4.5	4.2 (10 ppl)	Scale 1-5	User testing (Likert scale)
Aesthetic Appeal	3	4.5	4.5 (10 ppl)	Scale 1-5	User testing (Likert scale)
Deceleration smoothness	3	4.5	5 (Keys)	Scale 1-5	User testing (Likert scale)

As the table shows, our caliper design successfully met all of our metrics. However, some metrics were very close to their marginal limits. Since we technically fulfilled all of our metrics, for our final testing design we focused on improving the metrics that were close to the marginal values.

Key Insights from Initial Testing Process:

1. We experienced issues with caliper installation and assembly. We ended up needing to re-drill out our pivot screw holes. We originally created 6 mm diameter holes in our CAD model, but found that this actually resulted in a 5.9 mm hole due to overbuilding in 3D printing. So we decided to make the hole diameter 6.2 mm. This would allow for a clearance fit and the 0.1 mm "shrinkage" that we saw. Additionally, there was a tight amount of clearance for the bolt attaching to the brake cable. We need to extend the brake mounting hole out by at least a couple of mm to remedy this, and ensure that the screw head (10mm) is able to fit with some clearance.
2. Secondly, we observed deflection in our brake calipers. The maximum measured distance between the calipers at the brake cable mount was 12.73 mm. The distance between the calipers after the handbrake was pulled was 10.39 mm, meaning that the deflected amount was 12.73-10.39 = 2.34 mm. We predicted using FEA that the deflection would be 1.76 mm, meaning that the actual deflection was 22.8% higher than predicted. This was consistent with our understanding that FEA underestimates deflection. While we underestimated the deflection, our braking distance was still acceptable as it was right on the margin at 15.0m per ISO standards. Since the deflection and braking distance were just within our margins we decided to improve upon them with our final calipers. Additionally, our input force was measured at 120 N while the brakes were bottomed out. This meant to improve deflection and stopping distance while having a limited input force, we needed to make the calipers stiffer.
3. Finally, in testing our brakes, we noticed another important factor, which was the cost. Rounding up our mass value to 88 g, maximum mass given the uncertainty of our measurement, and using the given value of $0.20 per gram to print Nylon-12, the total cost of both calipers was $17.60 including material and manufacturing costs. So, reducing the mass of the caliper also decreases cost, which is an important design consideration.

Overall, through our first round of testing, we learned that our caliper did successfully meet every metric we outlined. Thus, we decided to focus the improvements on our next iteration on increasing caliper stiffness, adapting hole size and location for easier assembly, and trying to reduce mass if possible.

5. Design Improvements Between Tests

Overall Goals of Improvements:

Changes to the design between tests were predominantly aimed at improving two metrics: braking distance and ease of assembly. During our first round of testing, our recorded values fell right on the marginal values, which were a 15 m stopping distance and a 3 rating on a Likert scale for ease of assembly. Design improvements largely focused on addressing these metrics, with additional considerations for the effects of these changes on mass and aesthetics.

Tolerance and Assembly:
During our initial testing, we observed issues when assembling the brake calipers on the bike, specifically that we had to drill the pivot screw holes to create a clearance fit. Therefore, we improved upon our ease of assembly by increasing the pivot screw hole diameter from 6 mm to 6.2 mm to create a clearance fit, accounting for the overbuilding seen in SLS printing, and avoiding re-drilling. We also recorded that there was zero clearance between the brake cable mount screw in the right caliper and the side of the left caliper. This caused the bolt to roll up the side of the left caliper as tension was applied to the cable, restricting the freedom of the right caliper's displacements and making assembly difficult. To ensure free vertical movement of the screw head, we extended the brake mounting hole on the right caliper by 1 mm and adjusted widths of the left caliper, leading to a distance of 6.5 mm between the screw hole and left caliper as opposed to the previous 5 mm. Now the 5 mm radius screw has 1.5 mm of clearance as opposed to 0 mm. Together, these changes increased the ease of assembly.

Stiffness and Braking Distance:
To decrease braking distance, we knew our caliper design must increase the normal force applied at the brake pads, in order to decelerate quicker. We initially thought that increasing the mechanical advantage of our design, specifically by increasing r_{cable} the distance from cable mounting hole to pivot hole, would increase the normal force applied at the brake pad; this idea was rationalized using moment balance equations. However, this reasoning was incomplete, because we were incorrectly analyzing our calipers as rigid bodies. In reality, we needed to account for the effect of deflection and its effect on the translated force. $\frac{F_{in}}{F_{out}} \propto L$ but $\delta \propto L^3$, so increasing r_{cable} had a much larger impact on the deflection of the calipers. Therefore, we realized that since we are working with Nylon-12, which is a relatively flexible material, that increasing the stiffness of the part would have a much greater effect on increasing normal force that we see than the mechanical advantage.

To increase the stiffness of the calipers, we ensured that "bottoming out" the calipers during testing would result in a greater normal force on the brake wheels. This was achieved by increasing the thickness of the calipers and editing the cross-sections. The long cantilever beam section of the right caliper was updated from a cross-section of 17 x 20 mm to 20 x 25 mm. The bottom cross-section of the left caliper was changed from 17 x 16 mm to 18 x 15.5 mm, decreasing slightly to allow for clearance with assembly. We additionally adjusted the magnitudes of the input force and normal forces used in FEA required to stop the bike, by changing the designed stopping distance from 15m to 10m. Our goal was to shorten the stopping distance while maintaining or decreasing deflection, which would signal that the calipers could withstand larger forces with less yielding, so that more of the applied tension force would be translated to the normal force. Our updated tension forces for the right and left calipers were 289.88 N. The force of friction on the brake pad required to stop the bike was 260.89 N.

Minimizing Mass and Other Considerations:

Finally, we analyzed how to minimize mass while still maintaining the increase in stiffness achieved by increasing cross-sectional areas. To ensure this increase in mass was the most effective at reducing the bending moment, we created new cutouts from the middle of the right caliper to create an I-beam cross section, running FEA on versions with and without cutouts to compare deflection amounts and found negligible differences. We also increased the thickness of the cutouts in the left caliper. By removing material from our thickened caliper, we were able to minimize the addition of mass and thus the increased cost to print this part. Our mass increased between the two designs, but the redesign still met our ideal value set through benchmarking. In the original design, the left caliper was 37 g and the right was 50.5 g, totaling 87.5 g. In contrast, the new design has a left caliper of 32.5 g and a right caliper of 66 g, totaling 98.5 g.

Finally, we changed the curvature where the calipers fit together just to make them fit together more smoothly. This was done mostly for aesthetic reasons since the calipers would already have a washer between them to prevent additional frictional forces between the two caliper surfaces, but also to ensure they did not interfere with each other in use. A visual comparison of the initial testing designs and the final testing designs are shown in Figures 27 and 28 below.

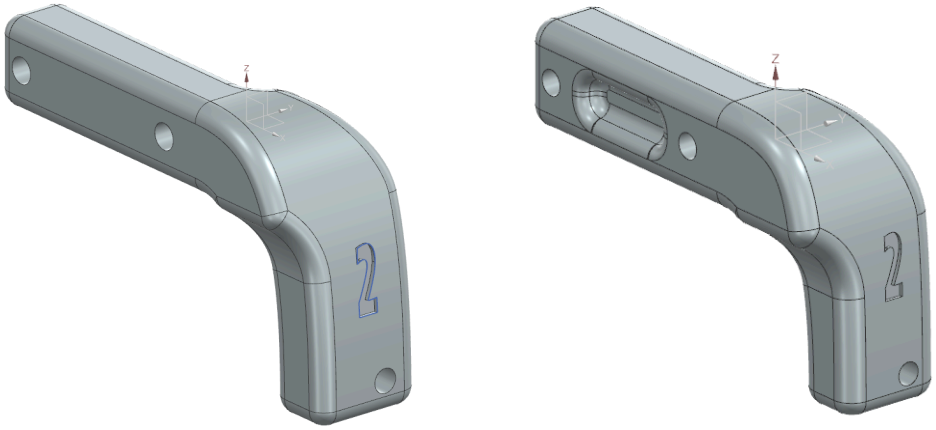


Figure 27: Side-by-side comparison of right caliper design used in initial testing and design used in final testing, with changes to cross-sectional area, cutouts, and fit of the assembly and hole tolerances.

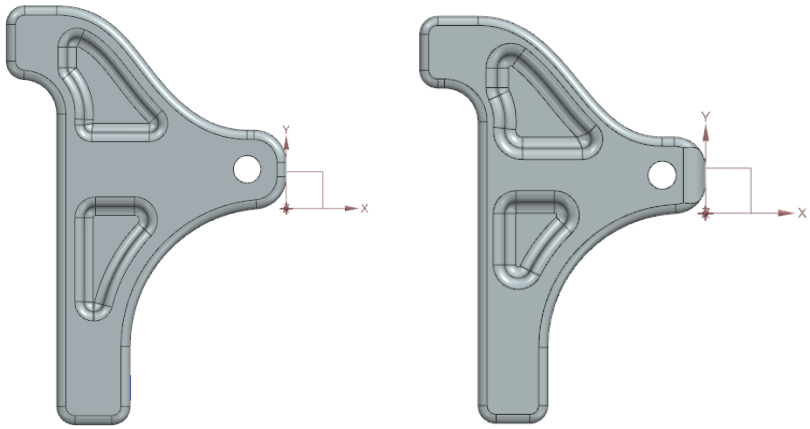


Figure 28: Side-by-side comparison of left caliper design used in initial testing and final testing, with changes to cross-sectional area, cutouts, fit of the assembly, and hole tolerances.

Comparison of FEA from Initial to Final Design:

We then compared and discussed the observed changes in stress and deflection values between the initial and final designs to predict changes in performance. For our FEA, we used our load values as calculated in Section 3.3. For the initial design, we used the marginal braking distance value of 15 m, which resulted in Tension = 198 N, and Friction due to the brake pad = 174 N. For the final design, we used our ideal braking distance value of 10 m, which resulted in Tension = 290 N, and Friction due to the brake pad = 261 N.

A comparison of the FEA between the initial and final design of the right caliper is shown in Figure 29 below. The relevant maximum stress value in the initial testing design is 21 MPa, compared to 18.88 MPa for the final testing design. Therefore we see the maximum stress decreased by 2.75 MPa in our current design, which is significant, especially considering the applied loads are 30% larger.

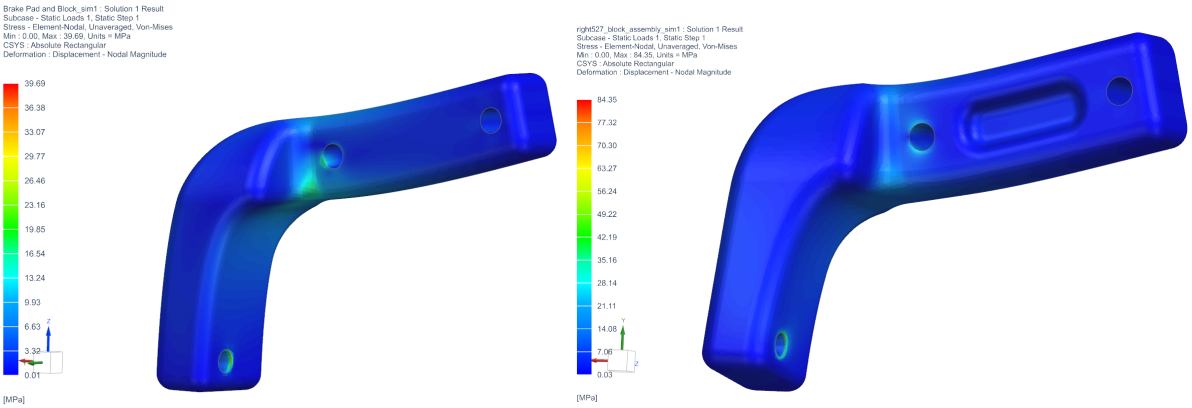


Figure 29: Comparison of Von-Mises stress of right caliper design from initial testing and final testing.

A comparison of the FEA between the initial and final design of the left caliper is shown in Figure 30 below. The relevant maximum stress value in the initial testing design is 21.63 MPa, compared to 27.64 MPa for the final testing design. The maximum stress value did increase, but we found this acceptable for multiple reasons. First, the applied forces in the FEA were increased by approximately 30%, which means it makes sense for the observed stresses to increase. Additionally, the observed stress value is still below our maximum allowable stress with a factor of safety of 1.3, which was deemed acceptable for our uses. Finally, the location of the highest observed stresses were in the corners of our cutouts, and we decided it was more important to include these to minimize the mass of our part than to reduce the maximum stress value.

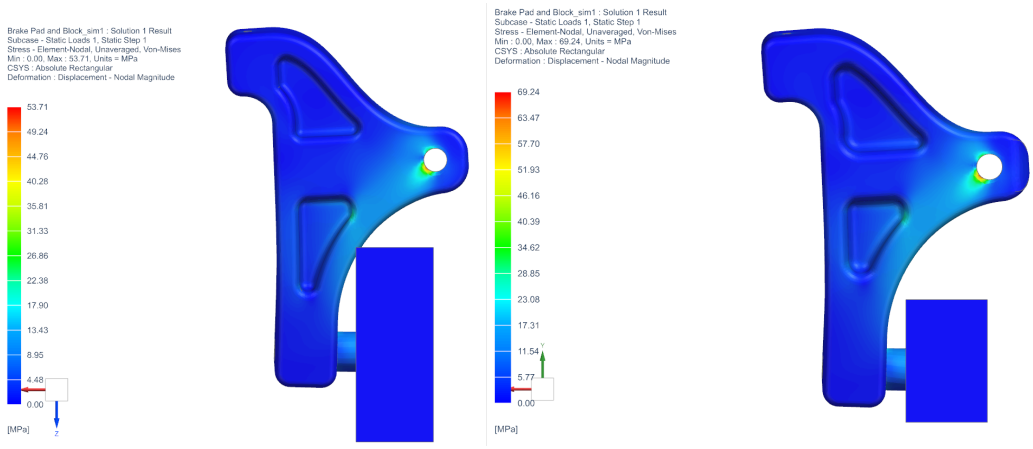


Figure 30: Comparison of Von-Mises stress of left caliper design from initial testing and final testing.

As calculated in Section 3.4, the total allowable deflection determined by our designed maximum reflection and factor of safety is 2.75mm. The total calculated deflection of our part for the initial testing design and final testing design are:
Initial Testing Design: 0.433 + 1.474 mm = 1.904 mm
Final Testing Design: 0.646 + 1.114 mm = 1.76 mm

Therefore, we see that the deflection has always met our maximum allowable deflection requirement with a factor of safety of 4, but that for our redesign the deflection has decreased by 7.5%, or by ~0.15 mm. This is again especially promising because these deflections were a result of 30% larger applied forces, which illustrate that we achieved our design goal of increasing the overall stiffness of our calipers.

6. Final Testing Results

In week 10, the final round of testing was conducted. Using the metric table initially constructed during the design process, it could be concluded that the calipers were a success. Every metric

was met, with most meeting the ideal value rather than the marginal value (8 out of 11), as illustrated in Table 8:

Color Key:

Did not meet metric	Met marginal value	Met ideal value

Table 8: Measurements of Metrics from Week 10 Testing of Calipers.

Metric	Marginal	Ideal	Measured Week 10	Units	Measurement Tool
Change in angular velocity before and after braking for 3 seconds	50% of initial angular velocity	0	0- braked for exactly 2.18 seconds	rpm	Video Freeze Frame
Braking distance (dry, initial velocity 25 km/hr)	15	10	12.2 m	meters	Measuring tape, Velocity meter, Equation 4 Lab 8 pdf
Mass	190	100	98.5 ± .5 g (66 g right, 32.5 g left)	grams	Scale
Input operating force	180	90	180 N (maximum applied force, bottomed out)	N	Force Gauge
Caliper Width	148	100	96.73 ± 0.01 mm	mm	Caliper
% change in brake-tire gap	10	0	0	%	User testing, by inspection
Cable Assembly (cable ends covered)	yes	yes	yes	Binary metric	User testing, by inspection
Ease of Assembly using basic tools	3	4	4 (Keys)	Scale 1-5	User testing (Likert scale)
Caliper's safety	3	4	4.02 (10 ppl)	Scale 1-5	User testing

based on feel & appearance					(Likert scale)
Aesthetic Appeal	3	4	3.73 (10 ppl)	Scale 1-5	User testing (Likert scale)
Deceleration smoothness	3	4	5 (Keys)	Scale 1-5	User testing (Likert scale)

Every design metric was met. Notably, we significantly decreased our braking distance from 15 m to 12.16m, which shows that our final design successfully increased the stiffness of the calipers. We were also able to increase our score for ease of assembly from a 3 to a 4 since we no longer had to re-drill holes and the new clearance for the brake cable screw allowed for easier installation and movement. During this testing process, however, it was important to note that the system now had new tubing and brake cable, there was possible wear on the brake pad from continual use, and the cable clamps were changed from brass to steel.

Key Insights from Final Testing Process:

1. The score for ease of assembly increased from a 3 to a 4 from initial testing, likely due to a perfect clearance fit with no drilling needed. This, however, was still not a 5 due to the design's multi-part assembly nature.
2. It was predicted that the deflection would be 1.904 mm using FEA, whereas the actual amount was 1.94 mm. This means that the actual deflection was 10.2% higher than the predicted one, which is consistent with the understanding that FEA underestimates deflection.
3. The braking distance decreased from 15 m in initial testing to 12.16 m, indicating the changes made to the design successfully increased the stiffness of the parts.
4. The input force increased to the maximum value at 180 N, from 120 N for initial testing.. More force was needed to deflect the calipers as the parts were now stiffer.
5. Mass increased in the re-design, but still remained below the ideal value.

7. Additional Improvements and Adjusting for Manufacturability

Improvements:

Based on the testing we completed in Week 10, we were able to meet all of the marginal values and many of the ideal values we set. However, there are still some possible improvements that could be made in future iterations of our caliper designs in order to further increase the marketability of our calipers. While we were able to decrease the braking distance from 15.0m to 12.2m over the course of two iterations of our designs, we were still not able to meet our ideal value of 10m. We attempted to decrease the braking distance by increasing the stiffness of our calipers, which we updated to have larger cross-sectional areas. This was not successful enough

to meet our ideal value, so future iterations may require further increases in the cross-section of the calipers. Additionally, we completed FEA on the calipers based on a new calculation of the input cable tension force to account for our goal of a 10m braking distance. In the future, we may want to choose an even shorter braking distance in our force calculations that would allow us a greater factor of safety.

The user testing we completed on the aesthetic appeal of our calipers showed an overall decrease in the visual aspect of our design. The testing we conducted was a survey of 10 classmates using a Likert scale. The lowered appeal value may be attributed to a small sample size, in which case future surveys should be designed to reach more people. Additionally, we may want to invest more time into market research to form a more visually appealing set of calipers.

We focused our attention on decreasing the stopping distance and increasing the ease of assembly of our calipers, which resulted in an overall increase in mass from the first to the second iteration. One of the objectives of our design was to be as lightweight as possible, which we were unable to strive for, due to our concentrated focus on meeting our marginal and ideal goals. We integrated some of the results from topology optimization into our caliper designs, namely, implementing recessed areas in the left caliper where stresses were simulated to be low. However, in the future, we could improve our mass minimization further by doing more robust and detailed optimization calculations and observations.

Manufacturing Scenarios:

Entry-Level Bike:

For an entry-level bike with a focus on cost efficiency and acceptable aesthetics, the main manufacturing technique used should be forming. Compared to casting and forging, forming has a relatively low setup and tooling cost while providing medium complexity and a shorter lead time. Additionally, forming is better suited to very high production volume compared to the other techniques. This bike should not use machining as the main manufacturing technique, as machining is not suitable for large-volume production. Additionally, machining would be very slow for deep, complex features that are present in the bike.

High-Performance Road Bike:

For a high-performance road bike that needs to be strong, lightweight, and have a premium finish, the manufacturers should consider using forging. Forging is most suitable for these goals because the high initial tooling investment is justified with high-volume production. It also gives a good feature tolerance and can deal with a good amount of complexity within the part. Additive manufacturing should not be used as it creates a rough surface finish that needs further post-processing to smoothen. CNC

Machining would also be more difficult to use as it is not as able to support large-volume production.

Custom Bike Frame Shop:

For a bike frame shop that manufactures custom bikes, the manufacturing technique must support flexibility, low tooling cost, and fast design iterations. This means that the shop should choose to use CNC machining over additive manufacturing. Comparing the two methods, CNC machining has a tight feature tolerance compared to the loose and medium tolerances given through additive manufacturing. The setup and tooling costs of both methods are low, and the lead time of both methods is also short. Both methods can also handle high part complexity. The distinguishing factor that puts CNC machining above additive manufacturing is the tight tolerance and higher precision allowed through the process.

8. Discussion

The overall design process for our brake calipers was challenging, yet rewarding. We were able to learn and utilize many important engineering techniques, such as finite element analysis and topology optimization software, to develop our final design. We completed initial modeling through NX CAD of our available design spaces for each caliper, then made iterative refinements based on our testing and simulation results. We applied engineering analysis knowledge like Castigliano's Method, to calculate the important values of forces and stresses that our brake calipers would experience under loading.

There were many useful aspects of the design process that we were able to experience through the development of our calipers. When conducting FEA through the NX software, we were able to predict the areas of high stress concentration and deflection that may otherwise have been unintuitive. We were also able to implement exact loads onto the system and isolate our calipers to determine the full effects of external forces. We used topology optimization, again through NX, in order to form a more mass-efficient design. The results we gained from the optimization allowed us to make improvements to the mass minimization aspect of our calipers without sacrificing performance. We learned about the tradeoff between the two parameters and made informed decisions about what we would prioritize within our own designs. The immediate feedback after simulating and testing our calipers was also very useful for us and made the iterative design process very smooth and efficient. We were able to quickly identify the strengths and weaknesses of each of our iterations and then proceed to make improvements based on our data.

We also faced several challenges in the designing of our brake calipers. One of the main issues that we encountered was that our simulation results deviated from reality. When we conducted

FEA on our initial brake caliper designs, then manufactured them using the SLS printing technique, we found that we had trouble assembling the components together. Due to the nature of additive manufacturing methods, we needed to calibrate our holes and fits over the course of two iterations. This cost us a lot of time that could have been allocated towards more focus on mass minimization. Additionally, the deflection results from the FEA simulation underestimated the true deflection that would occur with loading the physical part. This made it slightly more difficult to develop our calipers as we had to consider a greater factor of safety in our calculations. Another challenge we faced was balancing the trade-off between optimizing weight and maintaining strength within the calipers. We focused a lot on increasing the strength and stiffness of our calipers because we felt that it was more important to ensure a good braking distance before improving the weight. However, we ended with a final design that had increased in mass, which is the opposite of our desired result. Finding the balance between making lightweight and strong calipers is something that we struggled with and would need to put more effort into if we were to continue iterating on our designs.

Throughout this process, we learned the importance of using simulation and analysis softwares to guide mechanical design. We also gained plenty of experience with the iterative design process, going through multiple cycles of testing and refinement to reach our final design. Overall, this experience reinforced the idea that successful engineering design requires both technical analysis and practical problem-solving. It deepened our understanding of how analysis tools can enhance the quality of our designs, and emphasized the importance of adaptability, critical thinking, and iteration in bringing a concept to life.

Appendix A: Benchmarking for Price and Weight of Bike Calipers

Table 1 below shows price and weight benchmarking data, which was found by comparing different caliper models. These values were averaged to assist and used as the marginal values of our design metrics. The ideal values were created with the idea to further optimize these values, to ensure our caliper design was commercially viable and even better compared to actually available commercial options.

Table 1: Brake caliper price and weight benchmarking among commercially available products

Brake Caliper	Price ($)	Weight per caliper (g)	Link
R741 Caliper Set, Dual Pivot Rim	64.99	150	https://dealer.trpcycling.com/product/r741/
Shimano Ultegra BR-R8100 Rim Brake Calipers (Black)	81.5	181	https://www.performancebike.com/shimano-ultegra-brr8100-rim-brake-calipers-black-pair-ibrr8100a82/p1357208?v=1355229
SRAM Force Mechanical Brake Caliper	81	145	https://www.jensonusa.com/sram-force-axs-mechanical-brake-caliper-d1
SRAM Red AXS Mechanical Brake Caliper B2	183	131	https://www.jensonusa.com/sram-red-axs-mechanical-brake-caliper-d1
Shimano Alivio T4000 V Brake Caliper	25.74	188	https://www.chainreactioncycles.com/shimano-alivio-t4000-v-brake-caliper-931029#colcode=93102903
Average:	87.246	159	

Appendix B: Force Calculations

Force calculations for the tension force in the brake cable, the normal force from the wheel and the braking friction force from the wheel were calculated using a spreadsheet displayed below. The calculations were based upon our part's geometry as well as the testing velocity, testing mass, and subsequent maximum braking distance outlined in the ISO standards.

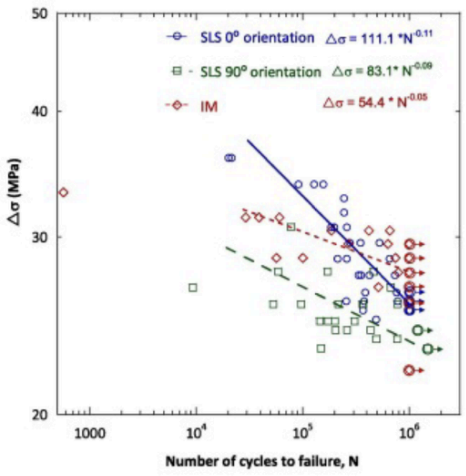
Table 1: Spreadsheet calculations of forces on calipers

ISOSTANDARDS (Don't Change)	
Mass of biker (kg)	100
Dry Velocity (km/hr)	25
Wet Velocity (km/hr)	16
Maximum Allowable Dry Braking Distance (m)	15
Maximum Allowable Wet braking Distance (m)	10
Bike Dimensions (Do not change)	
Radius of Wheel aka r_tire: (cm)	33
Dist. to brake pad aka r_brakepad: (cm)	30.5
Energy-Work Relationship for Braking distance and Velocity:	
F of friction ground for Dry: (N)	160.7510288
Force of friction ground for Wet: (N)	98.7654321
F friction on ground to F friction on brake pad (Moment Balance)	
F of Friction Brake for Dry: (N)	173.9273426
F of Friction Brake for Wet: (N)	106.8609593
Force of Friction to Normal Force Relationship:	
mu (dry):	0.9
Normal Force for Dry: (N)	193.2526029
Normal Force for Wet: (N)	118.7343992
Right Caliper:	
Distance from pivot to cable mount pt (r_cable): (mm)	46.5
Distance from pivot to brake pad mount pt (r_pad): (mm)	47.5
Left Caliper:	
Distance from pivot to cable mount pt (r_cable): (mm)	46.5
Distance from pivot to brake pad mount pt (r_pad): (mm)	47.5
Normal Force to Tension in Cable (Moment balance around pivot):	
	Right Caliper:
Tension for Dry: (N)	197.4085729
Tension for Wet: (N)	121.2878272
	Left Caliper:
Tension for Dry: (N)	197.4085729
Tension for Wet: (N)	121.2878272

Appendix C: Calculations for Endurance Strength at Cyclical Failure

We made estimations about the average use of a city bike and calculated use cycles based on said estimations. Loading cycles and material properties were both taken into account to then calculate endurance limit stress.

The below figure is courtesy of Salazar, et al. (2022; Engineering *Fracture Mechanics*), who performed reversed bending fatigue tests (see Lecture 7) on cylindrical SLS-Nylon-12 beams printed in differing orientations (shown in the two SLS graphs below; IM represents injection molded parts). *Note: Basquin's Law is also represented in Shigley's equation 6-13 on page 285.* $\Delta\sigma_{fl}$ *represents the endurance limit* (S_e).



Assuming basic city bike conditions:

- bike 30 mins a day
- brake once a minute

Assume material lifetime of ≈ 1 year due to weaker polymer

$30\ \frac{min}{day} \times 1\ \frac{brake}{min} \times 365\ \frac{day}{year} \times 1\ year = 10950\ brake\ cycles$

$N = 10{,}950$

Table 4. Basquin law ($\Delta\sigma = AN^b$) coefficients and fatigue limit at 10^6 cycles, $\Delta\sigma_{fl}$, of SLS PA-12 at 0° and 90° orientations [29] and of IM PA-12.

	Basquin law		$\Delta\sigma_{fl}$ (MPa)
	A	b	
IM	54.4	−0.05	28±1
SLS 0°	111.1	−0.11	26±1
SLS 90°	83.1	−0.09	23±1

Assuming 90° due to it being the weakest orientation (most conservative choice)

$\Delta\sigma = AN^b = 83.1(10950)^{-.09} = 36.0\ MPa$

Figure 1: Calculation of Lifetime of Brake Calipers